

JEE MAIN REVISION MASTER: WORK, ENERGY AND POWER

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Work-Energy Theorem & Conservative Forces	7 PYQs	Work & Energy
Moderate	Power & Collisions (Elastic & Inelastic)	3 PYQs	Collisions & Power

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Work-Energy Theorem & Conservative Forces	$W_{\text{net}} = \Delta K$, $W_{\text{ext}} = \Delta K + \Delta U$, $F = -\frac{dU}{dx}$, $U_{\text{spring}} = \frac{1}{2}kx^2$
Power & Collisions (Elastic & Inelastic)	$P = \vec{F} \cdot \vec{v} = \frac{dW}{dt}$, $e = \frac{v_2 - v_1}{u_1 - u_2}$, $\Delta K_{\text{loss}} = \frac{m_1 m_2}{2(m_1 + m_2)}(1 - e^2)(u_1 - u_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Work-Energy Theorem & Conservative Forces (Moderate Yield — 7 PYQs)

Core Principle: Work-energy theorem relating net work to change in kinetic energy, and potential energy gradients for conservative forces.

■■ **Caution / JEE Pitfall:** Work done by static friction on a pure-rolling body is strictly ZERO because the point of contact is instantaneously at rest.

■ **Speed Trick:** Stable equilibrium occurs where $(dU / dx) = 0$ and $(d^2U / dx^2) > 0$ (local potential minimum).

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change i..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "Two bodies with kinetic energies in the ratio of 3:1 are moving with equal linear momentum. The ratio of their masses is..." **[Ans: B]**
- **[2025 Shift] Q#3 (MCQ):** "A body of mass 0.25 kg travels along a straight line from $x=0$ to $x=2m$ with a speed $v=kx^{3/2}$ where $k=2SI$ units. The work done by..." **[Ans: N/A]**

3.2 Power & Collisions (Elastic & Inelastic) (Moderate Yield — 3 PYQs)

Core Principle: Instantaneous power delivery, linear momentum conservation, and coefficient of restitution in 1D and 2D impacts.

■ **Caution / JEE Pitfall:** In perfectly inelastic collisions ($e = 0$), momentum is conserved, but kinetic energy is NOT conserved.

■ **Speed Trick:** For two equal masses undergoing head-on elastic collision ($e = 1$), they completely exchange velocities.

Representative Source PYQs:

- **[2023 Shift] Q#1 (MCQ):** "A ball of mass 0.2kg is thrown vertically down from a height of 10m . It collides with the floor and loses 50% of its energy and then ..." [Ans: N/A]
- **[2020 Shift] Q#3 (MCQ):** "A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time t is propor..." [Ans: N/A]
- **[2017 Shift] Q#1 (MCQ):** "A motor pump lifts 6 tonnes of water from a well of depth 25m to the first floor of height 35m from the ground floor in 20 minutes. The po..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A horizontal force of 5N is applied on a stationary body of mass 5kg , which is initially at rest on a frictionless table. The change in kinetic energy of the body in 10s is

- (A) 25J
- (B) Zero
- (C) 125J
- (D) 250J

Correct Answer: None

Solution Analysis: The body is on a frictionless table, so the applied force produces acceleration without any loss due to friction. Given, $F=5\text{N}$ $m=5\text{kg}$ Using Newton's second law, $F=ma$ So, $a=(F/m)=(5/5)=1\text{m s}^{-2}$ The body starts from rest, so $u=0$ Time, $t=10\text{s}$ Using $v=u+at$ we get $v=0+1 \times 10=10\text{m s}^{-1}$ Initial kinetic energy is zero because the body is initially at rest. Final kinetic energy is $K=(1/2)mv^2$ $K=(1/2) \times 5 \times (10)^2$ $K=(1/2) \times 5 \times 100=250\text{J}$ So, the change in kinetic energy of the body is

Benchmark Problem 2: KCET 2023 (Q#1)

A ball of mass 0.2kg is thrown vertically down from a height of 10m . It collides with the floor and loses 50% of its energy and then rises back to the same height. The value of its initial velocity is

- (A) Zero
- (B) 14ms^{-1}
- (C) 196ms^{-1}
- (D) 20ms^{-1}

Correct Answer: None

Solution Analysis: Given, mass of the ball, $m=0.2\text{kg}$ Height from the surface of floor, $h=10\text{m}$ Let initial velocity be u . Total energy as initial point $TE=(1/2)mv^2+mgh$ After collision, remaining energy $=\frac{1}{2}mu^2+mgh$ With this remaining energy the ball bounces upto height h Therefore, $\frac{1}{2}mu^2+mgh=2mgh \Rightarrow (1/2)mu^2+mgh=2mgh \Rightarrow u^2=\sqrt{2gh}=\sqrt{2 \times 10 \times 10}=14\text{m/s}$